

Hemanth Reddy Nalabolu

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Hyderabad, India

SUMMARY

Applied AI Systems Engineer specializing in agentic AI infrastructure, evaluation, and reliability engineering. Built production voice-AI and interview platforms and an open-source agent reliability lab, turning statistical measurement rigor into shipped engineering decisions.

EXPERIENCE

AI Engineer

Hyderabad, India

AIRA HR – NR Consulting

August 2025 – Present

- Joined as an Intern and converted to full-time AI Engineer after 11 months; now own architecture for the company's AI Interview and AI Telephony platforms
- Re-architected a Selenium-based browser pipeline into native LiveKit; now powers 1,000+ production interviews and growing
- Built a full-duplex streaming voice pipeline (Deepgram STT, LLM inference, Cartesia TTS, LiveKit) minimizing end-to-end conversational latency
- Delivered an AI telephony platform with deterministic orchestration and RAG-based conversational intelligence, handling 2,500+ production calls and growing

SELECTED PROJECTS

Aegis — Agent Reliability & Evaluation Platform [\[GitHub\]](#)

Sep 2026

Python, FastAPI, PostgreSQL, MCP, MLflow

- Built an open-source reliability lab for agentic AI: versioned agents run against reproducible datasets with full tool/MCP/RAG trajectory capture
- Engineered a runtime-owned control-flow agent (property-tested state machine, budget/loop enforcement) behind an MCP gateway with deny-by-default policy
- Shipped statistically rigorous evaluation (McNemar, Benjamini–Hochberg, Wilson intervals) and 300+ tests; found and fixed a silent Postgres RLS tenancy bypass

MedAssist X — Serving-Time Admission Control [\[GitHub\]](#)

Jul 2025 – Sep 2026

Python, FastAPI, BM25 + Dense Retrieval, NumPy

- Built a multi-agent clinical assistant with an in-band release gate deciding, per request before any token is shown, whether an answer may reach the user
- Designed a six-check verification cascade (cheapest-first) catching dosage errors in 0.3ms that LLM judges routinely miss, with fail-closed ABSTAIN semantics
- Delivered 8,200+ lines and 448 tests running in 2.4s, including confidence calibration and a 15-attack red-team suite across 10 classes

Anomaly Transformer — Reimplementation & Evaluation [\[GitHub\]](#)

Oct 2024 – Sep 2026

Python, PyTorch, scikit-learn, Streamlit

- Reverse-engineered the ICLR 2022 Anomaly Transformer paper into a correctness-audited reimplementation, fixing 7 bugs in the loss and data pipeline
- Built a statistically rigorous evaluation layer (point-adjusted & PA%K F1, bootstrap CI) exposing 10x F1 inflation from evaluation convention alone, and benchmarked against 5 baselines across 28 machines with 68 tests

TECHNICAL SKILLS

Languages: Python, SQL, Java

AI / ML: PyTorch, Transformers, LangChain, LangGraph, RAG, Hybrid Retrieval, PEFT/LoRA/QLoRA

AI Evaluation: LLM-as-Judge Calibration, Statistical Testing (McNemar, BH-FDR), Red Teaming, Drift Detection

Agents & Backend: MCP, Agent Runtimes, FastAPI, AsyncIO, WebSockets, Redis, RBAC

Voice AI: LiveKit, WebRTC, Deepgram, Cartesia, Streaming AI

Data & Infra: PostgreSQL (RLS), MongoDB, FAISS, Docker, Git, GitHub Actions CI, MLflow, OpenTelemetry

EDUCATION

Neil Gogte Institute Of Technology

Hyderabad, India

B.E. in Computer Science and Engineering (CGPA: 8.07/10.0)

2022 – 2026